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SOFTWARE-DEFINED ANTENNA MANAGEMENT FOR AN ACCESS POINT OF A WIRELESS LOCAL AREA NETWORK

Abstract

Provided herein are techniques to facilitate software-defined antenna management for a wireless access point of a wireless local area network. In one example, a method may include communicating, by a wireless access point of a wireless local area network to a management service, a request to operate a transmission beam to be produced by the wireless access point using a transmitter and a software-defined antenna of the wireless access point; and obtaining by the wireless access point from the management service, beam configuration information identifying parameters that the wireless access point is to utilize for operation of the transmission beam to be produced by the wireless access point using the transmitter and the software-defined antenna.

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Background/Summary

TECHNICAL FIELD
[0001] The present disclosure relates to network equipment and services.
BACKGROUND
[0002] Networking architectures have grown increasingly complex in communications environments, particularly wireless networking environments. For wireless local area networks, the introduction of new frequency bands, such as the 6 Gigahertz (GHz) frequency band, presents new challenges and opportunities with regard to network management, including the management of wireless local area network access points.
[0003] In particular, outdoor wireless settings often include incumbent wireless transmit/receive devices, such as point-to-point microwave devices operating in certain Unlicensed National Information Instructure (U-NII) 6 GHz frequencies. Regulatory requirements stipulate that wireless local area network access points operating in the 6 GHz band are to limit interference to such incumbent devices operating in certain U-NII 6 GHz frequencies. Accordingly, challenges exist with regard to managing access points of wireless local area networks that operate in the 6 GHz band, particular in outdoor wireless environments in which incumbent wireless devices are operating.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] FIGS. **1**A and **1**B are diagrams illustrating example outdoor environments in which an access point of a wireless local area network may be operating in proximity to incumbent wireless devices operating in the 6 Gigahertz (GHz) band.
[0005] FIG. **2**A is a block diagram of a system that may be implemented to facilitate software-defined antenna management for a wireless an access point (AP) of a wireless local area network (WLAN), according to an example embodiment.
[0006] FIG. **2**B illustrates details associated with vertical orientations that can be provided by the software-defined antenna of the wireless AP of FIG. **2**A, according to an example embodiment.
[0007] FIG. **2**C illustrates details associated with horizontal orientations that can be provided by the software-defined antenna of the wireless AP of FIG. **2**A, according to an example embodiment.
[0008] FIG. **3** is a flow chart depicting a method according to an example embodiment.
[0009] FIG. **4** illustrates a hardware block diagram of a computing device configured to perform functions associated with operations discussed in connection with embodiments herein.
DETAILED DESCRIPTION
Overview
[0010] Innovations in wireless access points used in wireless local area networks (WLANs) have resulted in the development of directional antennas, also referred to herein as software-defined antennas. For a wireless access point (AP) configured with a software-defined antenna the transmission pattern or beam width for transmission beams transmitted or radiated via a transmitter and the software-defined antenna can be (automatically) switched between various configurations based on software or logic configured for the wireless AP. Such an innovation can be useful in many wireless environments, such as legacy wireless environments that may include APs having fixed-direction antennas.
[0011] A further innovation in wireless APs is utilization of the 6 Gigahertz (GHz) frequency band, which includes 5.925 GHZ-7.125 GHz frequencies, in which use of the 6 GHz band may facilitate increased throughput for wireless communications, among other improvements. Operation in the 6 GHz band also presents challenged, however, particularly in outdoor settings that can include incumbent wireless transmit/receive devices, such as point-to-point (p2p) microwave devices operating in certain U-NII 6 GHz frequencies.

[0012] Federal regulatory requirements stipulate that WLAN APs operating in the 6 GHz band are to limit interference to such incumbent devices operating in certain Unlicensed National Information Instructure (U-NII) 6 GHz bands/frequencies through power restrictions, such as Effective (or Equivalent) Isotropic Radiated Power (EIRP) restrictions, and orientation restrictions. For example, Federal Communication Commission (FCC) 14-30 Section 15.407 stipulates that for an outdoor wireless access point operating in the U-NII-1, U-NII-5, and U-NII-7 bands, "The maximum EIRP at any elevation angle above 30 degrees as measured from the horizon must not exceed 125 milliwatts (mW) (21 dBm)."
[0013] Current WLAN standards, such as Institute of Electrical and Electronics Engineers (IEEE) 802.11ax (Wi-Fi® 6/6E) and IEEE 802.be (Wi-Fi® 7) standards, do not include provisions that enable a wireless AP that is capable of software-defined antenna operation to communicate its beam capabilities (e.g., supported beam widths, supported orientations, and supported transmission power) to a control or management service (e.g., an automated frequency coordination (AFC) service, a wireless local area network (LAN) controller, or the like) to allow the control service to manage one of more beam configurations of the software defined antenna of the AP, for example, in order to reduce or limit interference to incumbent wireless devices operating in the 6 GHz band in outdoor environments (e.g., parking lots, stadiums, arenas, fields, etc.).
[0014] Embodiments herein provide techniques through which a wireless AP can provide beam information (for a transmission beam that can be operated via a software-defined antenna of the AP) to a control or management service, such as an AFC service, and can obtain beam configuration from the control service that may indicate one or more parameters by which the wireless AP is to operate the transmission beam via the software-defined antenna. Such enhancements, which can be incorporated into Wi-Fi standards, may introduce elements through which to improve the functionality of antenna directionality in 6 GHz operating environments.
[0015] In at least one embodiment, a computer-implemented method is provided that may include communicating, by a wireless access point of a wireless local area network to a management service, a request to operate a transmission beam to be produced by the wireless access point using a transmitter and a software-defined antenna of the wireless access point; and obtaining by the wireless access point from the management service, beam configuration information identifying parameters that the wireless access point is to utilize for operation of the transmission beam to be produced by the wireless access point using the transmitter and the software-defined antenna.
Example Embodiments
[0016] In a WLAN, one or more wireless APs provide wireless Radio Frequency (RF) coverage over which one or more wireless devices (e.g., phones, wearable devices, tablets, etc.) can connect to the APs in order to connect to one or more data networks (e.g., the public Internet, an enterprise network operated by an enterprise entity (e.g., a business, institution, university, etc.)), and/or the like.
[0017] In outdoor environments in which incumbent wireless devices operating in the 6 GHZ band may be present, steering for software-defined antennas for access points may be needed to limit interference with such incumbent device, per FCC regulations.
[0018] For example, consider FIGS. **1**A and **1**B, which illustrates example outdoor environments **100** (FIG. **1**A) and **100′** (FIG. **1**B) in which two incumbent microwave devices, **102-1** and **102-2**, are communicating via corresponding point-to-point (p2p) microwave links involving microwave transmissions in the 6 GHz band. For example, incumbent microwave device **102-1** can communicate with incumbent microwave device **102-1** via a first microwave link **104-1** operated at a first frequency in the 6 GHz band, shown in FIGS. **1**A and **1**B as 'f**1**'. Further, incumbent microwave device **102-2** can communicate with incumbent microwave device **102-2** via a second microwave link **104-2** operated at a second frequency in the 6 GHz band, shown in FIGS. **1**A and **1**B as 'f**2**'.
[0019] Further consider that a wireless AP **106** is present in outdoor environments **100**/**100′** and is

operated within a geographic proximity of the incumbent microwave devices **102-1** and **102-2**. Further consider that wireless AP **106** is configured with a software-defined antenna that can be configured provide an RF transmission beam that can be steered through a variety of beam patterns, configurations, or modes, including, but not limited to, a wide beam mode, a narrow beam mode, or a narrow beam mode having a vertical orientation (e.g., tilt angle) of a number of degrees from a center or boresight orientation (0=) 0° relative to a horizon or center of the software-defined antenna (e.g., 10°, 20°, or any other angle from center that may be facilitated via software steering capabilities provided for the AP **106**).

[0020] As shown in the outdoor environment **100** of FIG. **1**A, when wireless AP **106** is operated in the wide beam mode, a wide transmission beam **107** transmitted/radiated by the wireless AP **106** interferes/impacts the first and second microwave links **104-1**/**104-2** of incumbent microwave devices **102-1**/**102-2**. However, as shown in the outdoor environment **100**′ of FIG. **1**B, when wireless AP **106** is operated in the narrow beam mode, a narrow transmission beam **108** transmitted/radiated by the wireless AP **106** does not interfere or impact the first and second microwave links **104-1**/**104-2** of incumbent microwave devices **102-1**/**102-2**.

[0021] Thus, as illustrated in FIGS. **1**A and **1**B, it is desirable to manage beam configurations for a wireless AP having a software-defined antenna in order to limit interference for incumbent wireless devices operated within a geographic proximity to the wireless AP. In particular, it would be advantageous for a control or management service for an outdoor WLAN, such as an AFC service, to know the antenna/beam direction (also referred to herein interchangeably as ‘orientation’), beam width (also sometimes referred to as ‘beamwidth’), and transmission power (e.g., EIRP) for one or more beams of the wireless AP capable of operation via one or more software-defined antennas of the wireless AP.

[0022] Knowing such antenna/beam information for a wireless AP can greatly impact how much energy the wireless AP may use in a particular direction without causing interference to incumbent wireless devices within proximity to the wireless AP. In some instances, for example, software-defined antenna beams can play a significant role in an AFC process performed by an AFC service, which can reject or allow a grant or allow certain frequencies in the U-NII 5 or 7 bands that may not impact nearby incumbents. Without knowledge of antenna/beam information that can be provided by a wireless AP, the AFC process performed by the AFC service could simply reject a grant even though the antenna of the wireless AP may be pointing in an opposite direction from incumbents and, thus, would not impact the incumbents.

[0023] In accordance with embodiments herein, techniques are provided through which a wireless AP can communicate to a control or management service, such as an AFC service, a request to operate a transmission beam to be produced by the wireless AP using a transmitter and a software-defined antenna of the wireless AP. In response to request, the wireless AP can obtain beam configuration information from the management service that identifies one or more parameters that the wireless AP is to utilize for operation of the transmission beam to be produced by the wireless AP using the transmitter and the software-defined antenna.

[0024] Through embodiments herein, current and/or future Wi-Fi standards (e.g., Wi-Fi **6**, Wi-Fi **7**, IEEE 802.11bn (Wi-Fi **8**), and/or any other future Wi-Fi standards) can be improved/enhanced to enable an exchange of information between wireless APs operating software-defined antennas (sometimes referred to as ‘smart’ antennas) and a control or management service, such as an AFC service, in which the control service can provide beam configuration to the wireless APs in order to steer/switch beams operated by the wireless APs via software-defined antennas to different positions/orientations, modes, power levels, and/or frequencies

[0025] Referring to FIG. **2**A, FIG. **2**A is a block diagram of a system **200** that may be implemented to facilitate management of a software-software defined antenna for a wireless AP of a WLAN, according to an example embodiment. FIG. **2**B illustrates details associated with orientations of the software-defined antenna of the wireless AP of FIG. **2**A that can be managed for one or more transmission beams transmitted or radiated by the wireless AP, according to an example embodiment, and is discussed with reference to FIG. **2**A.

[0026] In at least one embodiment, system **200** may include a WLAN **210** that includes a wireless AP **220** and an automated frequency coordination (AFC) service **260** in which the wireless AP **220** and the AFC service **260** may interface with each other. In at least one embodiment, wireless AP **220** may include control logic **222**, orientation sensors **224**, an elevation sensor **226**, a memory **230**, and a transceiver **240** that may include a transmitter (Tx) **242** and a receiver (Rx) **244**. The wireless AP **220** may also include a software-defined antenna **250** (also referred to herein as a software-defined antenna assembly or simply an antenna). As shown in FIG. **2**A, the control logic **222** may interface with the orientation sensors **224**, the elevation sensor **226**, the memory **230**, the transceiver **240**, and the software-defined antenna **250**. The transceiver **240** may also interface with the software-defined antenna **250**.

[0027] AFC service **260** may include AFC logic **262** that interfaces with a memory **270** that can store incumbent device information **272** for one or more incumbent wireless devices that may be present within system **200**. For example, as shown in FIG. **2**A are incumbent wireless devices that can be present/located within a geographic proximity of WLAN **210**/wireless AP **220**, such as an incumbent wireless device **202-1** and an incumbent wireless device **202-2**. As illustrated in FIG. **2**A, incumbent wireless device **202-1** can communicate with incumbent wireless device **202-2** via an incumbent wireless link **204-1** operating at a first 6 GHz frequency, shown in FIG. **2**A as ‘f**1**’, and incumbent wireless device **202-2** can communication with incumbent wireless device **202-1** via an incumbent wireless link **204-2** operated at a second 6 GHz frequency, shown in FIG. **2**A as ‘f**2**’. In some example embodiments incumbent wireless devices **202-1** and **202-2** may interface with AFC service **260**. Some example embodiments, wireless AP **220** and incumbent wireless devices **202-1** and **202-1** may be provided in an outdoor environment (e.g., a parking lot, field, stadium, etc.).

[0028] The incumbent device information **272** stored via memory **270** may identify a geographic location for each of incumbent wireless devices **202-1** and **202-2**, such as vertical orientation of the devices (e.g., tilt angle), horizontal orientation of the devices (e.g., intercardinal/ordinal direction or angle of rotation (azimuth) over a horizontal plane relative to the horizon), transmission power of the devices, operating frequency or frequencies of the devices, and/or any other information that may be used to facilitate management of software-defined antennas for wireless APs (e.g., wireless AP **220**) in an environment. In various embodiments, the incumbent device information **272** can be configured for memory **270** by a network administrator, via an exchange with incumbent wireless devices **202-1** and **202-2**, via an exchange with a network controller, and/or using any other techniques as may be understood by a person having ordinary skill in the art.

[0029] Control logic **222** of wireless AP **220** can operate to perform functions associated with various operations discussed for embodiments herein and may store, maintain, or otherwise interact with information stored via memory **230** in order to facilitate operations for wireless AP **220** in accordance with embodiments herein.

[0030] Broadly, during operation of system **200**, wireless AP **220** can initiate a request (as generally shown at **280**) towards AFC service **260** for operation of a transmission beam to be produced by the wireless AP **220** within system **200**. The AFC service **260**, via AFC logic **262**, can perform an AFC process (as generally shown at **281**), based on the request and the incumbent device information **272** in order to determine beam configuration information that can be sent to the wireless AP **220** (as generally shown at **282**) in which the beam configuration information (shown in FIG. **2**A as beam configuration information **232**) is stored via memory **230** and used by the wireless AP **220** for configuration and operation of the transmission beam that is produced using the transmitter **242** and the software-defined antenna **250** in accordance with the beam configuration information **232**. Stated differently, the wireless AP **220**, via control logic **222** and orientation sensors **224**, can operate to configure at least one of the transmitter **242** and the software-defined antenna **250** to produce (transmit/radiate) the transmission beam by the wireless AP **220** in accordance with the beam configuration information **232**. Additional details regarding the AFC process (**281**) performed by the AFC service **260** are discussed in further detail herein. Generally, the AFC process performed by the AFC service **260** can operate to maximize spectrum availability for license-exempt devices (e.g., WLAN/Wi-Fi devices) through dynamic determination of channel availability at one or more locations in which the AFC process seeks to avoid/protect licensed operations in the 6 GHz frequency band (e.g., to avoid interfering incumbent wireless device **202-1**/**202-2** communications).

[0031] Thus, the beam configuration information **232** may identify parameters that the wireless AP **220** is to utilize to configure the transmitter **242** and/or the software-defined antenna **250** for one or more transmission beams to be produced by the wireless AP **220**. In various embodiments, the parameters identified in the beam configuration information **232** can identify any combination of inclusionary parameters that may be considered particular parameters that are to be/can be utilized/configured for the transmitter **242** and/or the software-defined antenna **250**) and/or exclusionary parameters that may be considered particular parameters that cannot be utilized/configured for the transmitter **242** and/or the software-defined antenna **250** (e.g., parameters/settings that are not to be used/are to be excluded from use by the wireless AP **220**.

[0032] Regarding operation of the wireless AP **220**, the orientation sensors **224** are configured to determine a vertical orientation (e.g., tilt angle, referred to herein as 'e') and a horizontal orientation (e.g., intercardinal/ordinal direction or angle of rotation (azimuth) over a horizontal plane relative to the horizon) of the software-defined antenna **250**. The control logic **222** is configured to estimate orientation information (vertical orientation and horizontal orientation) for the software-defined antenna using output from the orientation sensors **224**. The elevation sensor **226** is configured to measure an elevation (e.g., height above the ground/floor) of the wireless AP **220** (or software-defined antenna **250**) and the control logic is configured to determine an elevation of the wireless AP **220**/software-defined antenna **250** based on elevation output by the elevation sensor **226**.

[0033] The control logic **222** may further use orientation information from the orientation sensors **224** in order to control or manage the orientation of the software-defined antenna **250** based on the beam configuration information **232** obtained from the AFC service **260** for one or more transmission beams that the wireless AP **220** requests to operate. In at least one embodiment, the beam configuration may identify a vertical orientation (tilt angle) that the wireless AP **220**, via control logic **222**, is to utilize to control or manage (e.g., steer) the tilt angle of software-defined antenna **250**. In at least one embodiment, the beam configuration may identify, in addition to and/or in lieu of a vertical orientation, a horizontal orientation (azimuth) that wireless AP **220**, via control logic **222**, is to utilize to control or manage (e.g., steer) the horizontal direction/azimuth towards which the software-defined antenna **250** is pointed/directed. In accordance with embodiments herein, that software-defined antenna **250** may include, be configured with, and/or be controlled via any combination of motors, actuators, and/or any other control assembly/assemblies (not shown) that may facilitate controlling, managing and/or otherwise managing steering any combination of vertical orientation (tilt angle) and/or horizontal orientation (azimuth) of the software-defined antenna **250** via control logic **222**, beam configuration information **232**, and outputs from orientation sensors **224**. The control logic **222** is further configured to generate a beam transmission control to control or manage output power and transmission frequency of transmitter **242** for one or more transmission beams produced by the wireless AP based on the beam configuration information **232** obtained from the AFC service **260**.

[0034] In some example embodiments, the control logic **222** is configured to estimate an angle of tilt of the software-defined antenna **250** relative to the horizon or a center elevation of the software-defined antenna **250** based on the orientation determined by the orientation sensors **224**. In example embodiments, orientation sensors **224** may be implemented as a three-dimensional (XYZ) accelerometer that can detect the vertical orientation (tilt) of a device, such as software-defined antenna **250**, and a magnetometer that can detect the horizontal orientation or direction (intercardinal/ordinal direction or angle of rotation (azimuth) over a horizontal plane relative to the horizon) towards which the software-defined antenna is pointed/directed (e.g., for a particular beam to be transmitted by the wireless AP **220**). Even if an object, such as software-defined antenna **250**, is not moving the accelerometer can measure acceleration due to Earth's gravity, which is a constant downward force acting on all objects. Thus, the accelerometer can determine if the object is parallel to the Earth's surface or if it is tilted and, more specifically, can measure the tilt (in degrees).

[0035] Further regarding the orientation sensors **224**, the magnetometer can measure the direction towards which the software-defined antenna **250** is directed/pointed by measuring the Earth's magnetic field or magnetic moment.

[0036] Other sensors may be implemented for the orientation sensors **224** to measure vertical and/or horizontal orientations of the software-defined antenna, such as a gyroscope, an inertial measurement unit (IMU), or other similar sensors (now known or hereinafter developed).

[0037] Referring briefly to FIGS. **2**B and **2**C, FIG. **2**B illustrates details associated with vertical orientations (tilt angles) that can be provided by the software-defined antenna **250** of the wireless AP **220** for one or more transmission beams radiated or transmitted by the wireless AP **220**, according to an example embodiment, and FIG. **2**C illustrates details associated with horizontal orientations (direction/azimuth) that can be provided by the software-defined antenna **250** of the wireless AP of FIG. **2**A for one or more transmission beams radiated or transmitted by the wireless AP **220**, according to an example embodiment.

[0038] As shown in FIG. **2**B, which is a side view representation of the wireless AP **220**/software-defined antenna **250**, the software-defined antenna **250** may be controlled (by control logic **222**) in at least one instance to transmit or radiate a signal or transmission beam **252** having a beam width **254** (also sometimes referred to as transmit/radiation pattern) in a vertical direction in relation to the horizon. As also, shown in FIG. **2**B, the software-defined antenna **250** may have an elevation, relative to a center of the software-defined antenna, which can be measured by elevation sensor **226** (not shown). In the example of FIG. **2**B, the software-defined antenna **250** is controlled to transmit/radiate the transmission beam **252** in a vertical orientation (tilt angle, 0) of 0° relative to the horizon or the center elevation of the software-defined antenna **250**, which is assumed to be parallel with the horizon. In accordance with embodiments herein, the vertical orientation (tilt angle) of software-defined antenna can be controlled, via beam configuration information provided by AFC service **260**, to be set to a particular vertical orientation or tilt angle in which the tilt angle may be prescribed as a downward tilt angle relative to center (e.g., 10° downward from center, 20° downward from center, etc.). In some embodiments, a tilt angles can also be prescribed using upward tilt angles.

[0039] As shown in FIG. **2**C, which is a top-down view representation of the wireless AP **220**/software-defined antenna **250** the software-defined antenna **250** may have a horizontal orientation such that the transmission beam **252** can be transmitted in a particular direction for a horizontal angle, or azimuth, represented as ‘a’, relative to the cardinal North direction. In the example of FIG. **2**C, the software-defined antenna **250** is directed to transmit the transmission beam **252** for a horizontal orientation of 90° relative to the cardinal North direction. Other horizontal orientations for the software-defined antenna **250** can be envisioned. In some embodiments herein, the vertical orientation of software-defined antenna can be controlled, via beam configuration information provided by AFC service **260**, to be set to a particular horizontal orientation/direction or azimuth in which the azimuth may be prescribed as a negative/left/counterclockwise or positive/right/clockwise rotation relative to 0° or the cardinal North direction (e.g., −30° or 30° left/counterclockwise of 0°/North, 35° or 35° right/clockwise of 0°/North, etc.).

[0040] Returning to FIG. **2**A, the request/response (**280**/**282**) exchange between wireless AP **220** and the AFC service **260** can be performed in various forms and can encompass various AFC processes. It is noted that conventional AFC processes may not consider vertical/horizontal orientations (tilt/azimuth) for an antenna, however, embodiments herein envision that contention for the 6 GHz band may increase over time such that future AFC processes under future standards may consider any combination of vertical (tilt angle) and/or horizontal (azimuth) antenna information.
[0041] As noted above, during operation of system **200**, the wireless AP **220** (or a proxy thereof) may initiate a request towards the AFC service **260** for a grant to operate a transmission beam (also referred to interchangeably herein as a 'beam request') to which the AFC service **260** can provide a response (also referred to interchangeably herein as a 'beam response'), following an AFC process performed via AFC logic **262**, which may consider at least the elevation, horizontal orientation (azimuth), and/or beam switching of the software-defined antenna **250**, as well as the incumbent device information **272** in the AFC process. For the AFC process, the AFC logic **262** can, for example, determine whether there will be any potential impacts to incumbent wireless links **204-1** and **204-2** for one or more different potential configurations of a beam to be operated via the software-defined antenna **250** (e.g., wide, narrow, etc.).
[0042] In various embodiments, the request sent by the wireless AP **220** and the response provided by the AFC service **260** could be facilitated through an exchange involving spectrum inquiry request/response message or through a new request Information Element (IE) or message and a new response IE or message. In at least one embodiment, request/response IEs/messages may be provided as Vendor Specific Extension IEs/messages. In at least one embodiment, the request/response IEs/messages may be provided as standards-defined IEs/messages.
[0043] Different requests (beam requests) can be sent by the wireless AP **220** in accordance with embodiments herein. In some embodiments, a request sent by the wireless AP **220** may be a request for multiple grants involving multiple beams to be operated by the wireless AP **220**.
[0044] In some example embodiments, the wireless AP **220** may send the AFC service **260** a request that identifies the elevation of the software-defined antenna **250** and the horizontal orientation (azimuth) for the direction that the software-defined antenna **250** is pointed/directed. In such embodiments, the AFC service **260**/AFC logic **262** can perform an AFC process using a coordination algorithm that considers at least the elevation and horizontal orientation (azimuth) of the software-defined antenna **250** and the incumbent device information in order to provide a response to the wireless AP **220** including beam configuration information **232**. The beam configuration information **232** may identify any combination of a beam width, a vertical orientation (tilt angle), a transmission power (EIRP level), and/or a transmission frequency that the wireless AP **220**, via control logic **222**, is to use to configure any combination of the software-defined antenna **250** and/or the transmitter **242** to produce (e.g., transmit/radiate) a transmission beam or beam(s) (which may or may not be the transmission beam(s) requested by the wireless AP **220**, for example, if the transmission beam(s) requested by the wireless AP **220** is/are not allowed or is/are adjusted in some manner/characteristic by the AFC process).
[0045] In various embodiments, beam configuration information **232** sent to the wireless AP **220** can be identified using enumerated values, flags, Attribute-Value Pairs (AVPs), and/or any other mechanism through which beam configuration information **232** (e.g., wide transmission beam, narrow transmission beam, narrow with angle 10°, narrow with angle 20°, potentially horizontal orientation/azimuth, EIRP level or value, frequency, etc.) can be identified and used by the wireless AP **220** to configure any combination of the software-defined antenna **250** and/or the transmitter to operate (e.g., transmit/radiate) a transmission beam.
[0046] Other request/response exchanges can be envisioned in accordance with embodiments herein. In some example embodiments, the wireless AP **220** may send the AFC service **260** a request for a grant for a particular beam configuration (or multiple particular beam configurations)

in which the request identifies the elevation and horizontal orientation of the software-defined antenna **250** and identifies a requested beam configuration (or multiple configurations) that may identify any combination of beam width, orientation (vertical and/or horizontal), transmission power, and/or transmission frequency for the transmission beam(s) that the wireless AP **220** seeks to transmit/radiate.

[0047] In some example embodiments, the wireless AP **220** may send the AFC service **260** a request for a grant in which the request identifies the elevation and horizontal orientation of the software-defined antenna **250** and further includes beam capability information that identifies transmission capabilities of the wireless AP **220**. In various embodiments, the beam capability information may identify any combination of a range of beam widths supported by the wireless AP **220**, a range of vertical orientations (tilt angles) supported by the wireless AP **220**, a range of horizontal orientations (azimuths) supported by the wireless AP **220**, a range of transmission powers supported/capable by the wireless AP **220**, and/or a range of transmission frequencies supported/capable by the wireless AP **220**.

[0048] Thus, in accordance with embodiments herein, a request sent by the wireless AP **220** to the AFC service **260** (or any control or management service) may include at least one of: an elevation of the software-defined antenna **250** of the wireless AP **220**; beam width information indicating at least one beam width (or beam width mode) for the transmission beam (or beams) that the wireless AP **220** requests to operate or is capable of operating; orientation information indicating at least one of a vertical orientation (tilt angle) and/or a horizontal orientation (azimuth) of the software-defined antenna **250** for the transmission beam (or beams) that the wireless AP **220** requests to operate or is capable of operating; frequency information indicating at least one transmission frequency for the transmission beam (or beams) that the wireless AP **220** requests to operate or is capable of operating; and/or power information indicating at least one transmission power for the transmission beam (or beams) that the wireless AP **220** requests to operate or is capable of operating.

[0049] Different responses (beam responses) including different beam configuration information **282** can be sent by the AFC service **260** in accordance with embodiments herein, based on different information included in a request received from the wireless AP **220** and/or based on different AFC processes/algorithms that may be performed via the AFC logic **262**.

[0050] Consider various AFC processes that may be performed by the AFC service **260**/AFC logic **262**. In some instances, for example, the AFC logic **262** may determine that if the wireless AP **220**/software-defined antenna **250** were to be operated as wide transmission beam, there could be a possible impact to incumbent wireless devices **202-1** and **202-2** in the frequencies f**1** and f**2** where f**1**, f**2** belong to U-NII 5 and 7 bands. In such instances, the AFC logic **262** may only assign grants to the wireless AP **220** to operate the software-defined antenna **250** for frequencies excluding f**1** and f**2**.

[0051] Thus, in some instances, the AFC service **260** may provide a response to wireless AP **220** that included beam configuration information that identifies transmission frequencies for beams that the wireless AP **220** may not operate within/or that are to be excluded from operation by the wireless AP **220**. Accordingly, in some example embodiments, the beam configuration information **232** sent to wireless AP **220** may be identify exclusionary parameters, such as excluded beam width(s), tilt angle(s), azimuth(s), transmission power(s), and transmission frequencies, that the wireless AP **220** is not allowed to utilize for one or more beam transmissions.

[0052] In some instances, AFC logic **262** may perform a more aggressive AFC process and determine that the wireless AP **220** may use f**1** and f**2**, but at lower power (EIRP) levels. In such instances, the beam configuration information **232** sent to the wireless AP **220** can identify the transmission frequencies and the transmission powers that the wireless AP **220** is to use to configure transmitter **242** for operation of one or more beams.

[0053] In still some instances, the AFC logic **262** may determine that the wireless AP **220** may be allowed to operate a narrow transmission beam at all U-NII 5 and 7 bands, which information (potentially including vertical and/or horizontal orientation information, transmission frequencies, and/or powers, etc.) can be identified via beam configuration information **232** sent to the wireless AP **220**. In still some instances, such bands may be excluded from operation at certain frequencies, or the wireless AP **220** may be allowed to operate within them at higher EIRP levels for a wide beam mode.

[0054] In still some instances, the AFC logic **262** may determine that a wide transmission beam is to be used at certain sites/locations based on potential impacts to incumbents. In still some instances, a wide transmission beam may be used at lower EIRP levels.

[0055] Thus, based on different configurations of the wireless AP (e.g., elevation, vertical/horizontal orientation, etc.), spectrum grants of frequency, EIRP may vary for different AFC processes that may be performed by the AFC service **260**/AFC logic **262** such that beam configuration information **232** can be included in a response (or responses) for each one or more transmission beams that the wireless AP **220** may seek to operate.

[0056] In accordance with embodiments herein, parameters of beam configuration information **232** sent to wireless AP **220** may identify one or more of: a beam width (or multiple beam widths) that the wireless AP **220** is to utilize (e.g., inclusionary parameters) or is not to utilize (e.g., exclusionary parameters) to configure the software-defined antenna **250** to operate/produce a transmission beam (or multiple beams); orientation information (vertical and/or horizontal) that the wireless AP **220** is to utilize or is not to utilize to configure the software-defined antenna **250** to operate/produce a transmission beam (or beams); at least one transmission frequency that the wireless AP **220** is to utilize or is not to utilize to configure the transmitter **242** to operate/produce the transmission beam (or beams); and/or at least one transmission power that the wireless AP **220** is to utilize or is not to utilize to configure the transmitter **242** to operate/produce the transmission beam (or beams). Thus, the wireless AP **220** may configure at least one of the transmitter **242** and/or the software-defined antenna **250** based on the parameters of the beam configuration information **232** and may operate/produce the transmission beam (or beams) by performing beam transmissions via the transmitter and the software-defined antenna.

[0057] Accordingly, embodiments herein may enhance the ability of the wireless AP **220**/control logic **222** to determine whether to use different beam widths and/or EIRP levels for different scenarios (e.g., wide transmission beam with lower EIRP levels or narrow transmission beam with higher EIRP levels), to use different vertical orientations (tilt angles) and/or horizontal orientations (azimuths) for different scenarios (e.g., that may provide a balance between beam width and EIRP levels), and/or the like.

[0058] Referring to FIG. **3**, FIG. **3** is a flow chart depicting a method **300**, according to an example embodiment. In at least one embodiment, method **300** illustrates operations that may be performed by at least one of a wireless AP, such as the wireless AP **220** as shown in FIG. **2**A, and a control or management service, such as the AFC service **260** as shown in FIG. **2**A, in order to facilitate software-defined antenna management for the wireless AP according to an example embodiment.

[0059] At **302**, the method may include communicating, by a wireless access point of a wireless local area network to a management service, a request to operate a transmission beam to be produced by the wireless access point using a transmitter and a software-defined antenna of the wireless access point. At **304**, the method may include obtaining by the wireless access point from the management service, beam configuration information identifying parameters that the wireless access point is to utilize for operation of the transmission beam to be produced by the wireless access point using the transmitter and the software-defined antenna.

[0060] Referring to FIG. **4**, FIG. **4** illustrates a hardware block diagram of a computing device **400** that may perform functions associated with operations discussed herein in connection with the techniques described for embodiments herein. In various embodiments, a computing device or apparatus, such as computing device **400** or any combination of computing devices **400**, may be configured as any entity/entities in order to perform operations of the various techniques discussed for embodiments herein, such as any elements, functions, etc. discussed for embodiments herein (e.g., wireless AP **220**, AFC service **260**, etc.).

[0061] In at least one embodiment, the computing device **400** may be any apparatus that may include one or more processor(s) **402**, one or more memory element(s) **404**, storage **406**, a bus **408**, one or more network processor unit(s) **430** interconnected with one or more network input/output (I/O) interface(s) **432**, one or more I/O interface(s) **416**, and control logic **420**. In various embodiments, instructions associated with logic for computing device **400** can overlap in any manner and are not limited to the specific allocation of instructions and/or operations described herein.

[0062] For embodiments in which computing device **400** may be implemented as any device capable of wireless communications, computing device **400** may further include at least one baseband processor or modem **410**, one or more radio RF transceiver(s) **412** (e.g., any combination of RF receiver(s) and RF transmitter(s)), one or more antenna(s) or antenna array(s) **414** (which may be inclusive of software-defined antenna(s) or antenna array(s) in accordance with embodiments herein.

[0063] In at least one embodiment, processor(s) **402** is/are at least one hardware processor configured to execute various tasks, operations and/or functions for computing device **400** as described herein according to software and/or instructions configured for computing device **400**. Processor(s) **402** (e.g., a hardware processor) can execute any type of instructions associated with data to achieve the operations detailed herein. In one example, processor(s) **402** can transform an element or an article (e.g., data, information) from one state or thing to another state or thing. Any of potential processing elements, microprocessors, digital signal processor, baseband signal processor, modem, PHY, controllers, systems, managers, logic, and/or machines described herein can be construed as being encompassed within the broad term ‘processor’.

[0064] In at least one embodiment, memory element(s) **404** and/or storage **406** is/are configured to store data, information, software, and/or instructions associated with computing device **400**, and/or logic configured for memory element(s) **404** and/or storage **406**. For example, any logic described herein (e.g., control logic **420**) can, in various embodiments, be stored for computing device **400** using any combination of memory element(s) **404** and/or storage **406**. Note that in some embodiments, storage **406** can be consolidated with memory element(s) **404** (or vice versa) or can overlap/exist in any other suitable manner.

[0065] In at least one embodiment, bus **408** can be configured as an interface that enables one or more elements of computing device **400** to communicate in order to exchange information and/or data. Bus **408** can be implemented with any architecture designed for passing control, data and/or information between processors, memory elements/storage, peripheral devices, and/or any other hardware and/or software components that may be configured for computing device **400**. In at least one embodiment, bus **408** may be implemented as a fast kernel-hosted interconnect, potentially using shared memory between processes (e.g., logic), which can enable efficient communication paths between the processes.

[0066] In various embodiments, network processor unit(s) **430** may enable communication between computing device **400** and other systems, entities, etc., via network I/O interface(s) **432** (wired and/or wireless) to facilitate operations discussed for various embodiments described herein. In various embodiments, network processor unit(s) **430** can be configured as a combination of hardware and/or software, such as one or more Ethernet driver(s) and/or controller(s) or interface cards, Fibre Channel (e.g., optical) driver(s) and/or controller(s), wireless receivers/transmitters/transceivers, baseband processor(s)/modem(s), and/or other similar network interface driver(s) and/or controller(s) now known or hereafter developed to enable communications between computing device **400** and other systems, entities, etc. to facilitate operations for various embodiments described herein. In various embodiments, network I/O interface(s) **432** can be configured as one or more Ethernet port(s), Fibre Channel ports, any other I/O port(s), and/or antenna(s)/antenna array(s) now known or hereafter developed. Thus, the network processor unit(s) **430** and/or network I/O interface(s) **432** may include suitable interfaces for receiving, transmitting, and/or otherwise communicating data and/or information (wired and/or wirelessly) in a network environment.

[0067] I/O interface(s) **416** allow for input and output of data and/or information with other entities that may be connected to computing device **400**. For example, I/O interface(s) **416** may provide a connection to external devices such as a keyboard, keypad, a touch screen, and/or any other suitable input and/or output device now known or hereafter developed. In some instances, external devices can also include portable computer readable (non-transitory) storage media such as database systems, thumb drives, portable optical or magnetic disks, and memory cards. In still some instances, external devices can be a mechanism to display data to a user, such as, for example, a computer monitor, a display screen, or the like.

[0068] For embodiments in which computing device **400** is implemented as a wireless device or any apparatus capable of wireless communications, the RF transceiver(s) **412** may perform RF transmission and RF reception of wireless signals via antenna(s)/antenna array(s) **414**, and the baseband processor or modem **410** performs baseband modulation and demodulation, etc. associated with such signals to enable wireless communications for computing device **400**.

[0069] In various embodiments, control logic **420** can include instructions that, when executed, cause processor(s) **402** to perform operations, which can include, but not be limited to, providing overall control operations of computing device; interacting with other entities, systems, etc. described herein; maintaining and/or interacting with stored data, information, parameters, etc. (e.g., memory element(s), storage, data structures, databases, tables, etc.); combinations thereof; and/or the like to facilitate various operations for embodiments described herein.

[0070] The programs described herein (e.g., control logic **420**) may be identified based upon application(s) for which they are implemented in a specific embodiment. However, it should be appreciated that any particular program nomenclature herein is used merely for convenience; thus, embodiments herein should not be limited to use(s) solely described in any specific application(s) identified and/or implied by such nomenclature.

[0071] In various embodiments, any entity or apparatus as described herein may store data/information in any suitable volatile and/or non-volatile memory item (e.g., magnetic hard disk drive, solid state hard drive, semiconductor storage device, random access memory (RAM), read only memory (ROM), erasable programmable read only memory (EPROM), application specific integrated circuit (ASIC), etc.), software, logic (fixed logic, hardware logic, programmable logic, analog logic, digital logic), hardware, and/or in any other suitable component, device, element, and/or object as may be appropriate. Any of the memory items discussed herein should be construed as being encompassed within the broad term ‘memory element’. Data/information being tracked and/or sent to one or more entities as discussed herein could be provided in any database, table, register, list, cache, storage, and/or storage structure: all of which can be referenced at any suitable timeframe. Any such storage options may also be included within the broad term ‘memory element’ as used herein.

[0072] Note that in certain example implementations, operations as set forth herein may be implemented by logic encoded in one or more tangible media that is capable of storing instructions and/or digital information and may be inclusive of non-transitory tangible media and/or non-transitory computer readable storage media (e.g., embedded logic provided in: an ASIC, digital signal processing (DSP) instructions, software [potentially inclusive of object code and source code], etc.) for execution by one or more processor(s), and/or other similar machine, etc. Generally, memory element(s) **404** and/or storage **406** can store data, software, code, instructions (e.g., processor instructions), logic, parameters, combinations thereof, and/or the like used for operations described herein. This includes memory element(s) **404** and/or storage **406** being able to store data, software, code, instructions (e.g., processor instructions), logic, parameters, combinations thereof, or the like that are executed to carry out operations in accordance with teachings of the present disclosure.

[0073] In some instances, software of the present embodiments may be available via a non-transitory computer useable medium (e.g., magnetic or optical mediums, magneto-optic mediums, CD-ROM, DVD, memory devices, etc.) of a stationary or portable program product apparatus, downloadable file(s), file wrapper(s), object(s), package(s), container(s), and/or the like. In some instances, non-transitory computer readable storage media may also be removable. For example, a removable hard drive may be used for memory/storage in some implementations. Other examples may include optical and magnetic disks, thumb drives, and smart cards that can be inserted and/or otherwise connected to a computing device for transfer onto another computer readable storage medium.

[0074] In one form, a computer-implemented method is provided that may include communicating, by a wireless access point of a wireless local area network to a management service, a request to operate a transmission beam to be produced by the wireless access point using a transmitter and a software-defined antenna of the wireless access point; and obtaining by the wireless access point from the management service, beam configuration information identifying parameters that the wireless access point is to utilize for operation of the transmission beam to be produced by the wireless access point using the transmitter and the software-defined antenna. In a least one instance, the transmission beam is operated at a transmission frequency in a 6 Gigahertz (GHz) band.

[0075] In at least one instance, the parameters of the beam configuration information identify one or more of: a beam width that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; orientation information that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; at least one transmission frequency that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam; or at least one transmission power that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam. In at least one instance, the beam width identifies a wide transmission beam width or a narrow transmission beam width that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam.

[0076] In at least one instance, the orientation information indicates at least one of a horizontal orientation that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; or a vertical orientation that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam. In at least one instance, the at least one transmission frequency is at least one frequency within a 6 Gigahertz (GHz) frequency band that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam. In at least one instance, the at least one transmission power is at least one Equivalent Isotropic Radiated Power (EIRP) level that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam.

[0077] In at least one instance, the method may further include configuring at least one of the transmitter or the software-defined antenna based on the parameters of the beam configuration information; and operating the transmission beam by performing transmissions via the transmitter and the software-defined antenna.

[0078] In at least one instance, the beam configuration information is provided to the wireless access point based on one or more incumbent wireless transmission devices operating within a geographic proximity of the wireless access point. In at least one instance, the beam configuration information is provided to the wireless access point for an automated frequency coordination

(AFC) process provided by the management service.

[0079] In at least one instance, the request includes at least one of: an elevation of the software-defined antenna of the wireless access point; beam width information indicating at least one width for the transmission beam that the wireless access point requests to operate or that the wireless access point is capable of operating; orientation information indicating at least one of a vertical orientation or a horizontal orientation of the software-defined antenna for the transmission beam that the wireless access point requests to operate or that the wireless access point is capable of operating; frequency information indicating at least one transmission frequency for the transmission beam that the wireless access point requests to operate or that the wireless access point is capable of operating; or power information indicating at least one transmission power for the transmission beam that the wireless access point requests to operate or that the wireless access point is capable of operating.

[0080] Accordingly, embodiments herein may broadly provide for the ability for a wireless AP to communicate the direction of its software-defined antenna (or antennas), its beamwidth, and/or any other information related to the operation of one or more transmission beams produced or to be produced by the wireless AP (e.g., elevation, frequency, power, etc.) to a control or management service such as AFC service. The wireless AP can receive beam configuration information from the control or management service that the wireless AP can use to operate/produce one or more transmission beams using its software-defined antenna (or antennas) in accordance with the received beam configuration information. Thus, enhancements/features discussed for embodiments herein may be incorporated into WLAN standards to introduce new elements that may be used to improve the functionality of directionality of 6 GHz operation for one or more wireless APs for different environments, such as outdoor environments.

Variations and Implementations

[0081] Embodiments described herein may include one or more networks, which can represent a series of points and/or network elements of interconnected communication paths for receiving and/or transmitting messages (e.g., packets of information) that propagate through the one or more networks. These network elements offer communicative interfaces that facilitate communications between the network elements. A network can include any number of hardware and/or software elements coupled to (and in communication with) each other through a communication medium. Such networks can include, but are not limited to, any local area network (LAN), virtual LAN (VLAN), wide area network (WAN) (e.g., the Internet), software defined WAN (SD-WAN), wireless local area (WLA) access network, wireless wide area (WWA) access network, metropolitan area network (MAN), Intranet, Extranet, virtual private network (VPN), Low Power Network (LPN), Low Power Wide Area Network (LPWAN), Machine to Machine (M2M) network, Internet of Things (IoT) network, Ethernet network/switching system, any other appropriate architecture and/or system that facilitates communications in a network environment, and/or any suitable combination thereof.

[0082] Networks through which communications propagate can use any suitable technologies for communications including wireless communications (e.g., 4G/5G/nG, IEEE 802.11 (e.g., Wi-Fi®/Wi-Fi6®), IEEE 802.16 (e.g., Worldwide Interoperability for Microwave Access (WiMAX)), Radio-Frequency Identification (RFID), Near Field Communication (NFC), Bluetooth™, mm.wave, Ultra-Wideband (UWB), etc.), and/or wired communications (e.g., T1 lines, T3 lines, digital subscriber lines (DSL), Ethernet, Fibre Channel, etc.). Generally, any suitable means of communications may be used such as electric, sound, light, infrared, and/or radio to facilitate communications through one or more networks in accordance with embodiments herein. Communications, interactions, operations, etc. as discussed for various embodiments described herein may be performed among entities that may directly or indirectly connected utilizing any algorithms, communication protocols, interfaces, etc. (proprietary and/or non-proprietary) that allow for the exchange of data and/or information.

[0083] In various example implementations, any entity or apparatus for various embodiments described herein can encompass network elements (which can include virtualized network elements, functions, etc.) such as, for example, network appliances, forwarders, routers, servers, switches, gateways, bridges, loadbalancers, firewalls, processors, modules, radio receivers/transmitters, or any other suitable device, component, element, or object operable to exchange information that facilitates or otherwise helps to facilitate various operations in a network environment as described for various embodiments herein. Note that with the examples provided herein, interaction may be described in terms of one, two, three, or four entities. However, this has been done for purposes of clarity, simplicity and example only. The examples provided should not limit the scope or inhibit the broad teachings of systems, networks, etc. described herein as potentially applied to a myriad of other architectures.
[0084] Communications in a network environment can be referred to herein as ‘messages’, ‘messaging’, ‘signaling’, ‘data’, ‘content’, ‘objects’, ‘requests’, ‘queries’, ‘responses’, ‘replies’, etc. which may be inclusive of packets. As referred to herein and in the claims, the term ‘packet’ may be used in a generic sense to include packets, frames, segments, datagrams, and/or any other generic units that may be used to transmit communications in a network environment. Generally, a packet is a formatted unit of data that can contain control or routing information (e.g., source and destination address, source and destination port, etc.) and data, which is also sometimes referred to as a ‘payload’, ‘data payload’, and variations thereof. In some embodiments, control or routing information, management information, or the like can be included in packet fields, such as within header(s) and/or trailer(s) of packets. Internet Protocol (IP) addresses discussed herein and, in the claims, can include any IP version 4 (IPv4) and/or IP version 6 (IPv6) addresses.
[0085] To the extent that embodiments presented herein relate to the storage of data, the embodiments may employ any number of any conventional or other databases, data stores or storage structures (e.g., files, databases, data structures, data or other repositories, etc.) to store information.
[0086] Note that in this Specification, references to various features (e.g., elements, structures, nodes, modules, components, engines, logic, steps, operations, functions, characteristics, etc.) included in ‘one embodiment’, ‘example embodiment’, ‘an embodiment’, ‘another embodiment’, ‘certain embodiments’, ‘some embodiments’, ‘various embodiments’, ‘other embodiments’, ‘alternative embodiment’, and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments. Note also that a module, engine, client, controller, function, service, logic or the like as used herein in this Specification, can be inclusive of an executable file comprising instructions that can be understood and processed on a server, computer, processor, machine, compute node, combinations thereof, or the like and may further include library modules loaded during execution, object files, system files, hardware logic, software logic, or any other executable modules.
[0087] It is also noted that the operations and steps described with reference to the preceding figures illustrate only some of the possible scenarios that may be executed by one or more entities discussed herein. Some of these operations may be deleted or removed where appropriate, or these steps may be modified or changed considerably without departing from the scope of the presented concepts. In addition, the timing and sequence of these operations may be altered considerably and still achieve the results taught in this disclosure. The preceding operational flows have been offered for purposes of example and discussion. Substantial flexibility is provided by the embodiments in that any suitable arrangements, chronologies, configurations, and timing mechanisms may be provided without departing from the teachings of the discussed concepts.
[0088] As used herein, unless expressly stated to the contrary, use of the phrase ‘at least one of’, ‘one or more of’, ‘and/or’, variations thereof, or the like are open-ended expressions that are both conjunctive and disjunctive in operation for any and all possible combination of the associated

listed items. For example, each of the expressions ‘at least one of X, Y and Z’, ‘at least one of X, Y or Z’, ‘one or more of X, Y and Z’, ‘one or more of X, Y or Z’ and ‘X, Y and/or Z’ can mean any of the following: 1) X, but not Y and not Z; 2) Y, but not X and not Z; 3) Z, but not X and not Y; 4) X and Y, but not Z; 5) X and Z, but not Y; 6) Y and Z, but not X; or 7) X, Y, and Z.
[0089] Each example embodiment disclosed herein has been included to present one or more different features. However, all disclosed example embodiments are designed to work together as part of a single larger system or method. This disclosure explicitly envisions compound embodiments that combine multiple previously discussed features in different example embodiments into a single system or method.
[0090] Additionally, unless expressly stated to the contrary, the terms ‘first’, ‘second’, ‘third’, etc., are intended to distinguish the particular nouns they modify (e.g., element, condition, node, module, activity, operation, etc.). Unless expressly stated to the contrary, the use of these terms is not intended to indicate any type of order, rank, importance, temporal sequence, or hierarchy of the modified noun. For example, ‘first X’ and ‘second X’ are intended to designate two ‘X’ elements that are not necessarily limited by any order, rank, importance, temporal sequence, or hierarchy of the two elements. Further as referred to herein, ‘at least one of’ and ‘one or more of can be represented using the’ (s)′ nomenclature (e.g., one or more element(s)).
[0091] One or more advantages described herein are not meant to suggest that any one of the embodiments described herein necessarily provides all of the described advantages or that all the embodiments of the present disclosure necessarily provide any one of the described advantages. Numerous other changes, substitutions, variations, alterations, and/or modifications may be ascertained to one skilled in the art and it is intended that the present disclosure encompass all such changes, substitutions, variations, alterations, and/or modifications as falling within the scope of the appended claims.

Claims

1. A method comprising: communicating, by a wireless access point of a wireless local area network to a management service, a request to operate a transmission beam to be produced by the wireless access point using a transmitter and a software-defined antenna of the wireless access point; and obtaining by the wireless access point from the management service, beam configuration information identifying parameters that the wireless access point is to utilize for operation of the transmission beam to be produced by the wireless access point using the transmitter and the software-defined antenna.
2. The method of claim 1, wherein the transmission beam is operated at a transmission frequency in a 6 Gigahertz (GHz) band.
3. The method of claim 1, wherein the parameters of the beam configuration information identify one or more of: a beam width that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; orientation information that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; at least one transmission frequency that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam; or at least one transmission power that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam.
4. The method of claim 3, wherein the beam width identifies a wide transmission beam width or a narrow transmission beam width that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam.
5. The method of claim 3, wherein the orientation information indicates at least one of: a horizontal orientation that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; or a vertical orientation that the wireless access

point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam.
6. The method of claim 3, wherein the at least one transmission frequency is at least one frequency within a 6 Gigahertz (GHz) frequency band that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam.
7. The method of claim 3, wherein the at least one transmission power is at least one Equivalent Isotropic Radiated Power (EIRP) level that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam.
8. The method of claim 3, further comprising: configuring at least one of the transmitter or the software-defined antenna based on the parameters of the beam configuration information; and operating the transmission beam by performing transmissions via the transmitter and the software-defined antenna.
9. The method of claim 1, wherein the beam configuration information is provided to the wireless access point based on one or more incumbent wireless transmission devices operating within a geographic proximity of the wireless access point.
10. The method of claim 9, wherein the beam configuration information is provided to the wireless access point for an automated frequency coordination (AFC) process provided by the management service.
11. The method of claim 1, wherein the request includes at least one of: an elevation of the software-defined antenna of the wireless access point; beam width information indicating at least one width for the transmission beam that the wireless access point requests to operate or that the wireless access point is capable of operating; orientation information indicating at least one of a vertical orientation or a horizontal orientation of the software-defined antenna for the transmission beam that the wireless access point requests to operate or that the wireless access point is capable of operating; frequency information indicating at least one transmission frequency for the transmission beam that the wireless access point requests to operate or that the wireless access point is capable of operating; or power information indicating at least one transmission power for the transmission beam that the wireless access point requests to operate or that the wireless access point is capable of operating.
12. One or more non-transitory computer readable storage media encoded with instructions that, when executed by a processor, cause the processor to perform operations, comprising: communicating, by a wireless access point of a wireless local area network to a management service, a request to operate a transmission beam to be produced by the wireless access point using a transmitter and a software-defined antenna of the wireless access point; and obtaining by the wireless access point from the management service, beam configuration information identifying parameters that the wireless access point is to utilize for operation of the transmission beam to be produced by the wireless access point using the transmitter and the software-defined antenna.
13. The media of claim 12, wherein the parameters of the beam configuration information identify one or more of: a beam width that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; orientation information that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; at least one transmission frequency that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam; or at least one transmission power that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam.
14. The media of claim 13, further encoded with instructions that, when executed by a processor, cause the processor to perform further operations, comprising: configuring at least one of the transmitter or the software-defined antenna based on the parameters of the beam configuration information; and operating the transmission beam by performing transmissions via the transmitter and the software-defined antenna.

15. A wireless access point, comprising: at least one memory element for storing data; and at least one processor for executing instructions associated with the data, wherein executing the instructions causes the wireless access point to perform operations, comprising: communicating, by the wireless access point to a management service of a wireless local area network, a request to operate a transmission beam to be produced by the wireless access point using a transmitter and a software-defined antenna of the wireless access point; and obtaining by the wireless access point from the management service, beam configuration information identifying parameters that the wireless access point is to utilize for operation of the transmission beam to be produced by the wireless access point using the transmitter and the software-defined antenna.

16. The wireless access point of claim 15, wherein the parameters of the beam configuration information identify one or more of: a beam width that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; orientation information that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; at least one transmission frequency that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam; or at least one transmission power that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam.

17. The wireless access point of claim 16, wherein the beam width identifies a wide transmission beam width or a narrow transmission beam width that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam.

18. The wireless access point of claim 16, wherein the orientation information indicates at least one of: a horizontal orientation that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam; or a vertical orientation that the wireless access point is to utilize or is not to utilize to configure the software-defined antenna to operate the transmission beam.

19. The wireless access point of claim 16, wherein the at least one transmission frequency is at least one frequency within a 6 Gigahertz (GHz) frequency band that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam.

20. The wireless access point of claim 16, wherein the at least one transmission power is at least one Equivalent Isotropic Radiated Power (EIRP) level that the wireless access point is to utilize or is not to utilize to configure the transmitter to operate the transmission beam.